

Navraj Dhaliwal

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EDUCATION

California State University, Fresno

Master of Science in Computer Science

Fresno, CA

January 2026 – December 2026

California State University, Fresno

*Bachelor of Science in Computer Science; **Advanced Cybersecurity Certificate***

Fresno, CA

August 2021 – December 2025

RELEVANT COURSEWORK

Security & Networking: Computer Security, System and Cloud Security, Biometric Security, Computer Networking, Wireless Communications

Systems & CS Core: Operating Systems, Advanced Computer Organization, Database Systems, Algorithms and Data Structures

EXPERIENCE

Student Assistant – IT Help Desk

California State University, Fresno – Fresno State Library

Fresno, CA

2024 – Present

- Provided front-line technical support to library patrons and staff, resolving hardware, software, and connectivity issues.
- Diagnosed and troubleshoot network connectivity problems including Wi-Fi, VPN access, and DNS resolution on university systems.
- Assisted users with account access issues, password resets, and multi-factor authentication setup on campus systems.
- Maintained and imaged workstations, ensuring systems were patched and compliant with university IT policies.
- Escalated unresolved issues to senior IT staff and documented recurring problems to support process improvement.

SECURITY & SOFTWARE PROJECTS

Biometric Authentication System | *Python, SQL, OpenCV, NumPy, scikit-image, PostgreSQL*

2026

- Developed a biometric authentication system for identity verification using custom machine learning classifiers.
- Implemented k-NN and Gaussian Naive Bayes classifiers from scratch, analyzing security tradeoffs in classification accuracy including **false acceptance rate (FAR)** and **false rejection rate (FRR)**.
- Extracted Gabor filter-based texture features to improve robustness against lighting and orientation variation.
- Integrated a **PostgreSQL** database to securely store biometric feature vectors with access-controlled queries.
- Evaluated system reliability using precision, recall, and confusion matrices aligned with real-world security thresholds.

Network Traffic Analysis Lab | *Wireshark, NS3, TCP/IP, Linux*

2025

- Built and simulated network topologies using **NS3** to analyze routing behavior and protocol performance.
- Captured and analyzed live and simulated traffic with **Wireshark**, identifying anomalous packets and protocol misuse.
- Investigated TCP/IP handshake sequences, ARP behavior, and DNS resolution to understand common attack surfaces.
- Documented findings on potential vectors including port scanning, packet sniffing, and man-in-the-middle exposure.

Real-Time Breakout Rooms | *JavaScript, Python, FastAPI, Socket.IO, HTML/CSS*

2025

- Built a real-time web application with dynamic room management, emphasizing **secure authentication** and access control.
- Enforced session isolation to prevent unauthorized room access and implemented input validation to mitigate injection risks.
- Integrated **WebSocket** communication via Socket.IO with message validation to ensure data integrity in real time.
- Designed backend APIs with attention to session integrity, applying secure coding practices throughout development.

TECHNICAL SKILLS

Security & Networking: TCP/IP, VPN, Firewalls, IDS/IPS, Network Security, System Security, Authentication, Access Control, Secure Coding

Security Tools: Wireshark, Nmap, NS3, VirtualBox, Kali Linux

Languages: Python, C++, Java, JavaScript, C

Backend & APIs: FastAPI, Flask, REST APIs, WebSockets

Databases: SQL, PostgreSQL, MySQL

Developer Tools: Git, Docker, Linux, GCP

Certifications: Advanced Cybersecurity Certificate – California State University, Fresno